DPT INFORMATIQUE
COMPUTER SCIENCE AND INFORMATION TECHNOLOGY
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INSTITUT NATIONAL DES SCIENCES APPLIQUÉES LYON

Software Development

Grammar and Languages

IDENTIFICATION

CODE :	IF-4-LG
ECTS :	1.0

HOURS

Lectures :	6.0 h
Seminars :	4.0 h
Laboratory :	4.0 h
Project :	0.0 h
Teacher-student contact :	14.0 h
Personal work :	10.0 h
Total :	24.0 h

ASSESSMENT METHOD

Validation of the practical work during the session.
Final exam with individual evaluation.

TEACHING AIDS

Lecture notes.

TEACHING LANGUAGE

French

CONTACT

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AIMS

Introduction to methods and tools for (formal) language processing (language design, parsing and translation, and thus the multiples uses of grammars in computer science).

This module develops the skill "Analyse and transform a language" by enforcing the following capacities :
- Design, transform and interpret a formal grammar
- Implement a lexical analysis
- Implement a syntactic analysis (top-down and bottom-up)

It also participates to the skill "Designe the architecture of an object oriented sofware" thanks to the following capacities :
- Structure a software into packages and weakly coupled classes
- Use the Design Patterns

CONTENT

Course
1. Lexical parsing, finite automata
2. Top-down parsing
3. Bottom-up parsing
4. Attributed grammars

This course is completed with a 4h practical on parsing (LL(1), LR(0), SLR(1) et LALR(1)) and a practical of 4h during which a complete parser is implemented in C++.

BIBLIOGRAPHY

[1] AHO, SETHI, ULLMAN. Compilers : Principles, Techniques and Tools. Addison Wesley Pub.

PRE-REQUISITE

Software Development and algorithms, Graph Theory, Logic Programming, XML, C++

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